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Parametric 3D Modeling of a Customized Prosthetic Hand Finger for Additive Manufacturing

L. Nini*, A. Ceccarelli, N. L. Tagliamonte, L. Zollo, F. Taffoni

Abstract— **In this paper, a new method to design a custom finger for a hand prosthesis is proposed. The prosthetic finger is composed of a linkage mechanism and three shells that resemble the shape of human phalanges (acquired with a 3D scanner). The proposed method is based on a parametric Computer-Aided Design (CAD) that allows the 3D modeling of the finger according to the different anthropometric parameters derived from of the user. The 3D modeling process takes into account also manufacturing constraints related to the production technology. Additive manufacturing has been selected to reduce costs and allow the production of components with a complex geometry resembling the human finger. The method has been tested on an adult volunteer by developing a replica of the index. The compliance of the mechanism dimensions with the anthropometric measures has been verified and a FEM-based analysis has been performed to estimate mechanism robustness and deformations. A deviation analysis has been also carried out to evaluate the similarity in shape between the finger scan and 3D-modeled shells. Finally, the finger prototype has been manufactured using a Selective Laser Sintering printer and the trajectory of the fingertip has been compared with the human one.**

Index Terms — **Upper Limb Prosthesis, Additive Manufacturing, Custom Design, Parametric Design.**

I. INTRODUCTION

In 2017, the World Health Organization reported that around 0.5% of the world population requires prosthetic and orthotic devices due to the loss of a limb or a functional impairment [1]. Amputations are frequently caused by trauma but also by congenital limb deficiency, vascular pathologies, or cancer [2]. About 38.7% of the cases reported in [1] regards the upper limb: 19.6% unilateral, 19.1% bilateral. The impact of upper limb amputation on an individual's quality of life is contingent upon the amputation level. In case of a total hand amputation, as in trans-radial or trans-humeral amputations, a significant number of activities of daily living (ADLs) are impossible [3]. Robotic hand prostheses have the potential to restore some of the functionalities lost due to hand amputation. However, the abandonment rate of these devices remains high [4]–[7]. In a survey of 92 upper limb amputees, Datta et al [4] point out how the main reasons for prosthesis abandonment are in the level of functionality (i.e. the number of ADLs they allow to restore), in the level of cosmesis, and in the weight, frequently judged excessive by patients. Recent review works [5]–[7] identify costs, difficulties of prosthetic care, and lacking of prosthetic training as further reasons limiting the use of these devices.

All the Authors are with the Laboratory of Advanced Robotics and Human-Centred Technologies – CREO Lab, Universitá Campus Bio-Medico di Roma, Via Àlvaro Del Portillo 21, 00128 – Rome, Italy.

*Corresponding Author: l.nini@unicampus.it

Cordella and co-authors [6] define a list of user requirements for hand prostheses that may impact on their acceptability. The anthropomorphism of the device in terms of size, weight, shape, and colour in addition to the capacity of the prosthesis to partially enable ADLs are two requirements to be fulfilled. Unfortunately, currently commercial devices allow a reduced level of customization and they are still quite expensive [3].

Several approaches have been pursued to support the design of custom prosthetic hands. They can be grouped in: *i)* discrete sizing and *ii)* continuous scaling methods. In the first one, a set of different prosthesis sizes are predefined according to the population distribution [8], [9]. The prosthesis model is selected as the one closest to the anthropometric characteristics of the user. The second method can be implemented by using a uniform scaling of the 3D model [10]–[12] or by introducing a parametric model of the CAD to scale the size of the prosthesis according to specific rules (*i.e.*, non-uniform scaling) [13]–[15]. In both cases, the finger shape of the prosthesis is predefined and not related to the patient's morphology. 3D scanning technologies may represent a possible source of information to be integrated into the customization process, as already done for the production of orthoses [16].

Additive Manufacturing (AM) is a production technique that can be effectively used to develop low-cost robotic hand prostheses [17], [18]. It reduces the cost for the production of a few replicas of custom components as well as production time, even if it has a lower dimensional accuracy compared with subtractive manufacturing technologies [19]. The object is obtained by adding material: it is usually split into layers along the vertical direction and it is manufactured by releasing material, according to the geometry of the section. This allows an optimal material distribution obtaining lighter components with good mechanical properties [20]. As a consequence, the cosmesis of the prosthesis could be improved promoting its acceptability [21]. Fused Deposition Modeling (FDM) and Selective Laser Sintering (SLS) are the most used AM techniques to produce robotic hand prostheses [17]. FDM is based on the melting of a thermoplastic filament that is extruded from a nozzle; SLS worked with powder which is sintered layer by layer by a laser source in a heated chamber. In both cases, the object developed may require a post-processing session to remove the support material or the unsintered powder, respectively.

The objective of this paper is to propose a novel method for parametric 3D modeling of prosthetic hand fingers, tailored to the specific morphology of the residual hand, captured through 3D scanning, and subsequently developed through the use of AM technology.

II. MATERIALS AND METHOD

Robotic prosthetic hands developed with AM techniques [17], [22] may be classified as tendon-based or linkage-based depending on the strategies used to transmit forces to the fingers. Tendon-based devices are the most spread ones since they enable shape adaptivity with a compact design, but they exert low grasping forces. On the contrary, linkage-based devices are more bulky but exert a more efficient force transmission [23] and, with the proper use of compliant elements, they enable also shape adaptivity [24]. Solutions based on pneumatic or hydraulic actuators have not been taken into account due to the weight and encumbrance of components necessary for their effective use (e.g., compressor, pumps, etc.).

In this work, a finger composed of a six-bar linkage mechanism and of covering shells mimicking the patient's residual hand morphology is considered to develop custom prosthetic fingers for adult patients (i.e., for hand length (HL) $\geq$ 159 mm, the 5th percentile of woman hand length [25]). The proposed method (Fig. 1) is based on four main blocks. The mechanism design block sizes the mechanism according to a set of patient features; the 3D scanning block captures hand morphology; the Parametric 3D modeling block customizes the CAD of the mechanism and of the shells according to patient characteristics and constraints due to AM technology selected for production. Finally, the Manufacturing and Assembly block produces the custom prototype. This work focuses on the Parametric 3D Modeling and on the Manufacturing and Assembly blocks, but few details on the other blocks will be provided to better explain the entire process.

A. Mechanism Design and 3D Scanning

The *Mechanism Design* block allows the target six-bar linkage mechanism dimensioning (Fig. 2) through a kinematic synthesis based on the loop-closure equations [26]. The mechanism is composed of six links (frame, L_1, L_2, proximal (*Prox*), intermediate (*Int*), and distal (*Dist*) phalanges) and seven joints (circles in Fig. 2). The frame models the hand

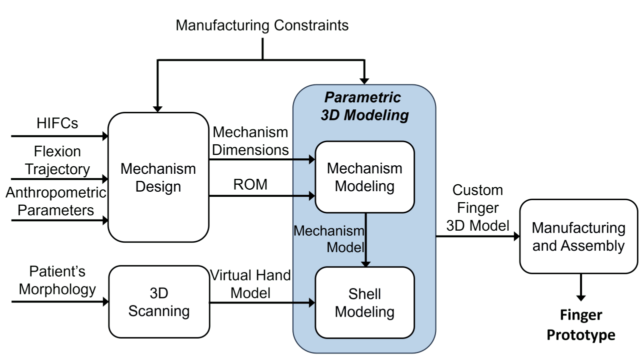

Fig. 1. Overview of the method proposed for the customization of prosthetic hand fingers. "HIFCs" stands for "Hand Intra-Finger Couplings" and "ROM" stands for "Range of Motion".

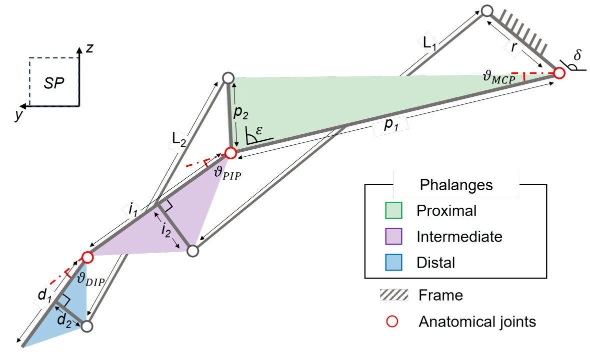

Fig. 2. Kinematic model of the six-bar linkage mechanism. Circles represent revolute joins and, in particular, red ones represent the MetaCarpoPhalangeal (MCP), Proximal InterPhalangeal (PIP), and Distal InterPhalangeal (DIP) joints; θ_{MCP},θ_{PIP}, and θ_{DIP} represent the joint angles. "SP" stands for "Sagittal Plane".

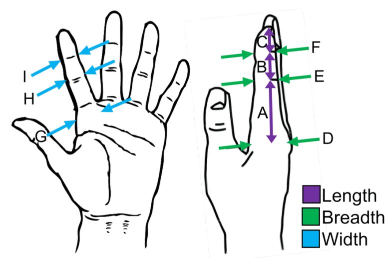

Fig. 3. Anthropometric parameters. Phalanges lengths: proximal (A), intermediate (B), and distal (C); Phalanges breadths at MCP (D), PIP (E), and DIP (F) joint level; and phalanx widths at MCP (G), PIP (H), and DIP (I) joint level.

palm whereas L_1 and L_2 are two links which connect the *frame* with the *int* and the *prox* with the *dist*, respectively. These last two links model the Hand Intra-Finger Couplings (HIFCs) among finger joints in free flexion movements [27] and close the kinematic chain. The potential solutions generated by the kinematic synthesis are filtered to identify those that are most compatible with the anthropometric parameters (Fig. 3), the HIFCs, and the finger flexion trajectory (in input to the mechanism design block). Overall, 6 features (see Fig. 2) are tuned to define the mechanism dimensions and its Range of Motion (ROM): 4 lengths (r, p_2, i_2, d_2) and two angles (δ, ε). The dimensions related to the phalanx lengths are dependent on the patient's anthropometric dimensions (p_1, i_1, d_1) while the length of the link L_1 and L_2 of the mechanism are defined as a consequence (Fig. 2). More details about the mechanism design block are described in [28].

3D scanning of the patient's residual hand is performed to acquire hand morphology. The patient is requested to sit in a comfortable posture and to place the hand on a specific support, designed to reduce as much as possible the deformation of finger soft tissues. A handheld scanner, Einscan HX model (SHINING 3D Tech Co., Ltd, China), is used to scan the hand of the patient with a single session, in rapid scan configuration (Blue LED light source). The scanner outputs a mesh file that is mirrored to replicate the missed hand. The virtual hand model is imported into the

CAD software and used as a reference to model the outer surface of the prosthetic finger shells.

B. Parametric 3D-Modeling

The mechanism dimensions and its ROM as well as the virtual hand model are input to the *Parametric 3D Modeling* block. This block aims to easily and quickly adapt the design of the prosthetic finger, in terms of mechanism and shells, to the features of different patients. For this reason, a parametric approach has been implemented in Fusion 360 CAD software (Autodesk Inc, USA). However, since AM techniques are selected for production, manufacturing constraints introduced by these technologies should be taken into account. Considering the dimensional accuracy of the FDM and SLS technologies (0.5 and 0.3 mm, respectively [19], [29]) as well as the necessity to manufacture irregular surfaces to resemble the anatomical ones, the SLS technology has been selected. It allows the manufacture of holes with a minimum diameter equal to 1.5 mm and walls with a minimum thickness of 0.8 mm. In particular, SLS printers working with Nylon 12 material (biocompatible) have been considered. The Fuse 1 3D printer (Formlabs Inc, USA) has been selected due to its printing chamber ($165 \times 165 \times 300$ mm^3) compatible with the development of a prosthetic hand. The anatomical reference frame is aligned with the CAD reference to match the sagittal, transversal, and coronal planes (SP, TP, and CP), with the *yz*-, *xz*-, and *xy*-planes of the CAD, respectively. A parametric sketch is defined on the SP based on the 11 mechanism dimensions and the initial joint angles output from the mechanism design block (Fig. 2). In detail, parameters are the mechanism dimensions defined during the kinematic synthesis (r, p_2, i_2, d_2, δ, ε, $L1$, $L2$), the proximal(p_1), intermediate (i_1) and distal (d_1) phalanx lengths in addition to the initial joint angles (θ^i_{MCP}, θ^i_{PIP}, and θ^i_{DIP}) obtained from the ROM. In particular, d_1 is the length of the patient's distal phalanx reduced by 4 mm to take into account the shell thickness at the fingertip (TIP) and the clearance between them and the mechanism model. To avoid a torsion effect during finger flexion, the structure has been designed to be symmetric to the SP. Links have been 3D modeled with a parametric extrusion along the x-axis. This extrusion takes into account: *i*) the phalanges width at MCP, PIP, and DIP joints (i.e., G, H, and I, respectively, see Fig 3). G is used to parameterize the frame width (b in Fig. 4), while H and I are used to verify that the final width of the mechanism is lower than the anthropometric measures, where the joints are positioned; *ii)* manufacturing constraints related to the clearance (cl), necessary to guarantee relative rotations among adjacent links (i.e. ≥ 0.2 mm), and the minimum thickness of walls ($w_{min} \geq 0.8$ mm); *iii)* a minimum thickness of link (l_{min}) that is set equal to 2.0 mm and verified with a FEM-based analysis; *iv)* the number of elements (n), that contribute to the width of the mechanism at the frame and joint levels as defined in Fig. 4. Table I reports in details the relations used for the parametric extrusion.

Joints are modelled as hinges and manufactured embedding commercial stainless steal pins into the mechanism links. Pins with a diameter (ϕ) of 2.0 mm have been chosen. This diameter ± 0.1 mm has been used to obtain clearance or interference fit to allow the rotation and to avoid axial movement, respectively. The center of each joint is defined on the parametric sketch. The modeled mechanism is tested on the ROM to verify the presence of interference among links and, eventually, to implement any necessary corrections. Finally, each link of the mechanism is individually tested with a FEM-based analysis on Autodesk NASTRAN software (Autodesk Inc, USA) to estimate robustness and deformations. A 1 N load is applied perpendicularly to the TIP of the mechanism in the most critical configuration (i.e., when the finger is extended in the initial configuration). A reaction torque of 0.15 Nm is considered at the MCP joint to simulate an actuator that balances the load on the TIP. Since the mechanism is described by a linear system, it is easy to evaluate reaction forces using a different load.

The target finger is manually cut from the virtual hand model and aligned with the mechanism along the longitudinal axis (i.e., the y-axis). The mechanism is used as a reference for the inner surface of the shells, while the scan of the target finger is used as a reference for the outer surface. In detail, shells are modeled using the parametric loft. This 3D modeling function allows the building of a solid between two boundary construction surfaces through a set of guidelines that drive the solid shape. Construction planes are defined at the joints and at the TIP level. The planes related to the MCP and DIP joints (planes a and c in Fig. 5) are perpendicular to p_1 and d_1, respectively. The plane associated with the

TABLE I
PARAMETERS DEFINITION

Parameter	Definition	Parameter	Definition
a	l_{min}	e	l_{min}
b	$\frac{0.75G}{2} - \frac{(n-1)}{2}cl - a$	f	$b - e - \frac{n-1}{2}cl$
c	$\frac{b}{2}$	g	$\frac{b-e-\frac{n-1}{2}cl}{2}$
d	$b - c - \frac{n-1}{2}cl$	h	$\frac{b-e-\frac{n-1}{2}cl}{2}$

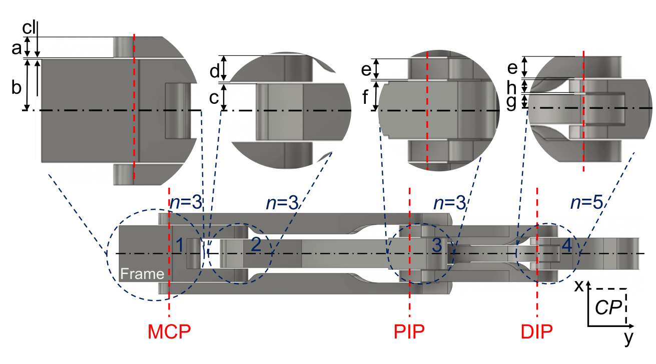

Fig. 4. Parametric extrusion of the mechanism.

PIP joint (plane b in Fig. 5) is parallel to the p_2 link and shifted distally to the joint to embed the entire Prox link of the mechanism into the shell. The TIP planes (planes d and e in Fig. 5) are both parallel to plane c and they are perpendicularly shifted of $d_1 + 2$ mm and $d_1 + 4$ mm, respectively. Two concentric parametric ellipses are defined on each construction plane. Vertices of the outer ellipses , defined as construction points (green points in Fig. 6) of the outer surface of the shell, fit with the anthropometric width and breadth of the finger at the corresponding joint level (i.e., G, H, I, and D, F, G, respectively, see Fig. 3). The inner ellipse is 2 mm smaller than the outer one and its vertices are the construction points of the inner surface of the shell. Other construction points are defined on d and e planes to close the inner and outer shell surfaces of the distal phalanx. As for the inner surface, construction points of different planes are connected with geometrical primitives (arcs or segments) paying attention to not overlap the primitive with the mechanism. On these primitives, a set of reference points (black points in Fig. 6) are defined and used to drive the internal guidelines for the loft function. Primitives of the outer surface are just segments connecting the outer construction points. Reference points (red points in Fig. 6) are defined in these segments. They can be manually moved along the direction perpendicular to the geometrical primitives to adapt the guidelines to the virtual finger model morphology. The position of these points tunes the external guidelines and changes the outer shape of the shells and their thickness, which must be higher than the minimum wall thickness allowed by the SLS technology (0.8 mm). After defining all the guidelines and construction surfaces, shells are modeled with the loft function by selecting the external ellipses, the point on plane e, and the external guidelines to create three new bodies. Material is removed from them through a cutting loft function defined on the internal ellipses, the point on the plane d and the internal guidelines (Fig. 6.c). Each shell is divided in two symmetric parts to be assembled on the corresponding phalanx with two M2 screws. Finally, the shells are virtually assembled on the mechanism model and the entire ROM is simulated in CAD environment. If interference among shells and/or between each shell and the mechanism is detected, the area of the shell responsible for the interference is manually removed to obtain the final 3D model of the custom finger.

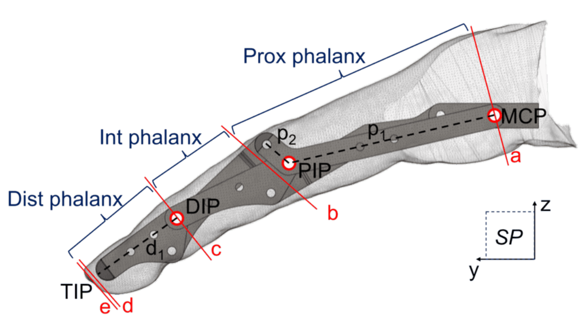

Fig. 5. Virtual finger and mechanism models alignment. Red lines represent the construction plane profiles.

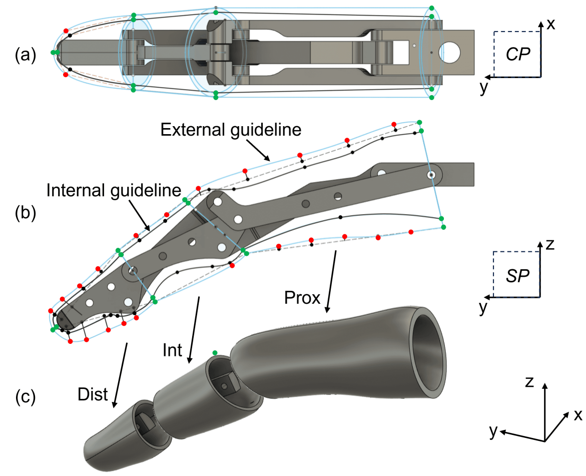

Fig. 6. Shell guidelines for loft function on the coronal plane (a) and on the sagittal plane (b). Green points represent the construction points; black points represent reference points of internal guidelines; red points represent the reference points of external guideline which could be moved perpendicularly to the outer geometrical primitives (dotted lines) to change the shell shape; (c) Three shells.

C. Experimental test

A healthy adult volunteer (female) aged 25 years, with $HL = 180.0$ mm has been recruited. The subject's hand morphology has been scanned to obtain the virtual hand model and the index has been selected (target virtual finger model) to test the method proposed. The compliance of the mechanism and the shells with the anthropometric parameters (Table II) and with the morphology of the volunteer is verified. In detail, the 3D model of the mechanism has been tested to verify: *i)* the compliance of parametric extrusion proposed in Table I with the subject anthropometric measures; *ii)* the compliance of estimated robustness and deformations of the structure with the material selected. Moreover, the outer surface of the shells is compared with the virtual finger model performing a deviation analysis between them in FreeCad software [30]. The virtual finger model is divided into three phalanges according to the planes a, b, c in Fig. 5 and exported as a single *.stl* file along with the shells in Fig. 6.c. Phalanges and shells (aligned with the same reference axis) are imported into the FreeCad software and the deviation among their outer surfaces is measured. Finally, the Custom finger 3D Model has been manufactured and assembled to obtain the finger prototype. The prototype has

TABLE II
VOLUNTEER'S ANTHROPOMETRIC PARAMETERS

	Length [mm]	Breadth [mm]	Width [mm]
MCP	$A = 44.0$	$D = 23.0$	$G = 20.0$
PIP	$B = 26.0$	$E = 18.0$	$H = 19.0$
DIP	$C = 18.0$	$F = 13.0$	$I = 17.0$

been flexed in controlled settings using a Universal testing machine and the TIP trajectory has been reconstructed using a open source video tracking software [31].

III. RESULTS AND DISCUSSION

Table III reports the extrusion parameters of the mechanism derived from the subject anthropometric dimensions according to the relations introduced in Table I.

TABLE III
CUSTOM MECHANISM MODEL: EXTRUSION PARAMETERS

Parameters	[mm]	Parameters	[mm]
a	2.0	e	2.0
b	5.3	f	3.1
c	2.65	g	1.45
d	2.45	h	1.45

The dimensions of the mechanism are compared with the subject's anthropometric measures. Comparison is carried out taking into consideration production constraints introduced by the method proposed. In details, phalanx lengths are set to be equal to the corresponding subject's measures. Link breadths range from a minimum thickness of 5.0 ± 0.1 mm (to guarantee a 1.5 mm thickness around holes for pins) to a maximum value set, at least, as subject's breadths reduced by the shell minimum thickness (i.e., 0.8mm). Mechanism widths (W) are estimated at joint levels as the sum of the width of links and clearances (Fig. 4):

$$W_{MCP} = W_{PIP} = 2(b + cl + a) = 15mm \quad (1)$$

$$W_{DIP} = 2(g + h + e + 2cl) = 10.6mm \quad (2)$$

W must be at least 1.6 mm narrower than the subject's finger widths: W_{MCP} is 5 mm narrower than the anthropometric dimension G (Fig. II), W_{PIP} is 4 mm narrower than H, and W_{DIP} is 6.4 mm narrower than I. Thus, the entire mechanism width remains into the anthropometric dimensions even when shells are assembled. The healthy subject's virtual finger model, the finger 3D model, and the prosthetic finger prototype are shown in Fig. 7. The 3D model (Fig. 7.b) and the prototype (Fig. 7.c) are flexed for the full ROM measured on the human subject. Their TIP trajectories are reconstructed (blue line and cyan dotted line respectively) and compared with the one measured on the human subject (red line). Root Mean Square Errors (RMSEs) between human vs. prototype or 3D model are respectively 1.00 mm and 0.89 mm [1]. The higher RMSE related to the prototype depend on the accuracy of the SLS printer used. The von Mises stress (σ) on the structure has been evaluated with the FEM-based analysis considering a perpendicular load applied to the TIP and an actuation torque at the MCP joint to balance the load (see section II). The maximum stress (σ_{max} = 14.0 MPa) is found on the Prox link at the MCP joint level (Fig. 8) and it is lower than the Yield stress of the Nylon 12 material ($\sigma_Y = 47.0$ MPa). The maximum stress on other links is everywhere lower than 5.5 MPa. Thus, the minimum thickness used to model links ($l_{min} = 2$ mm) guarantees the robustness of the mechanism when a load up to 3 N is applied on the TIP. The maximum displacement is observed in the Prox with a value of 0.2 mm. It reaches a maximum of 0.07 mm in other links.

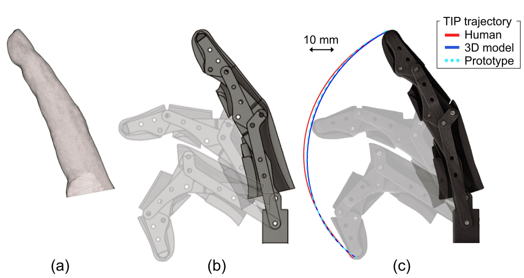

Fig. 7. Virtual finger model (a), finger 3D model (b), and finger prototype (c) based on a specific subject's anthropometric measures and morphology. The 3D model and the prototype are represented in three different configurations. The TIP trajectory of the subject, of the 3D model, and of the prototype are reported in (c).

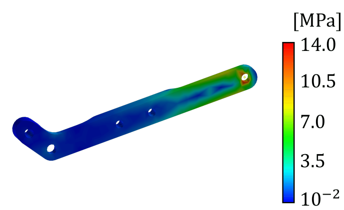

Fig. 8. FEM-based analysis results. The Von Mises stress is reported for the most stressed link, i.e, Prox.

The deviation analysis to test the similarity among shells and the virtual finger model is reported in (Fig. 9). A

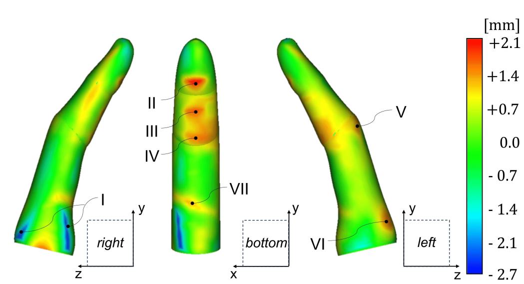

Fig. 9. Deviation analysis results.

negative/positive deviation means a shell smaller/greater than the model at that point. The minimum deviation is equal to −2.7 mm and it is located on the right of the proximal shell (point I in Fig. 9). This deviation depends on the ellipse used to model the finger geometry on the construction plane a (see Fig. 5) at the MCP level, which does not take into account the finger insertion into the palm. The maximum deviation is equal to 2.1 mm and it is identified on the bottom of the

[1] The demonstrative video is available at link https://www.youtube.com/watch?v=wpD3LcUUwdU

distal phalanx (point II in Fig. 9). It depends on the support structure developed to sustain the subject's hand during the 3D scanning that generates also a deviation of 1.7 mm on the bottom of the intermediate phalanx (point III in Fig. 9). Another deviation of 1.5 mm on the same phalanx (point IV in Fig. 9) is related to the mechanism encumbrances as well as the deviation on the top of the proximal phalanx (point V and VI in Fig. 9). In the end, a deviation of 1.2 mm is measured in correspondence with the digito-palmar crease (point VII in Fig. 9), which is not modeled by the shells to simplify their shape. Overall, the Root Mean Square (RMS) deviation is $+0.65$ mm.

The 3D modeling of a prosthetic finger using the method proposed takes about one hour due to interference check. Indeed, the removal of mechanical interference among shells and/or between each shell is operated manually and it should be made more automatic to reduce the modeling time. This preliminary study verifies the compliance of the prosthetic finger with the characteristics of the patient hand in terms of TIP motion trajectory and finger morphology. Further studies need to be carried out on patients to assess the effective user experience of the proposed device [6].

IV. CONCLUSIONS AND FUTURE WORKS

This work proposed a method for parametric 3D modeling of customized prosthetic hand fingers to be developed with AM technology and applied it to an adult volunteer. The method proposed, even if tested with SLS technology, can be applied also to other AM techniques simply changing the manufacturing constraints. This method could be applied also to the other fingers including the thumb. A future development will allow to exploit the method proposed for shells also to model the subject's palm to obtain a fully customized prosthetic hand. Moreover, by working on a simplification of the target mechanism for the finger, it will be possible to extend the production of custom prosthetic hands also to pediatric patients.

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